KNN Classification using Iris Dataset - Project Report

Abstract

This project implements a K-Nearest Neighbors (KNN) classification model using the Iris flower dataset. The workflow includes data loading, exploratory analysis, preprocessing, train-test split, model training, prediction, evaluation, and visualization.

Objective

To classify Iris flowers into Setosa, Versicolor, and Virginica species using the KNN algorithm and evaluate the model performance.

Dataset

The Iris dataset contains 150 samples with four numerical features: sepal length, sepal width, petal length, and petal width. The target variable is the flower species.

Methodology

1. Import required libraries.
2. Load the Iris dataset.
3. Explore and visualize the data.
4. Split the dataset into training and testing sets.
5. Train a KNN classifier.
6. Predict on the test data.
7. Evaluate using accuracy score, confusion matrix, and classification report.

Algorithm

KNN is a supervised learning algorithm that classifies a sample based on the majority class among its K nearest neighbors measured using a distance metric such as Euclidean distance.

Results

The notebook demonstrates successful model training and evaluation on the Iris dataset. The achieved accuracy depends on the selected value of K and train-test split, but typically exceeds 90% on this dataset.

Conclusion

KNN is an effective and simple classification algorithm for small structured datasets like Iris. Proper selection of K and feature scaling can further improve performance.

Tools & Libraries

Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Jupyter Notebook.